ConsensusCourt: Procedural Evidence Arbitration for Scientific Claims in Multi-Agent Research Pipelines

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Abstract

Multi-agent LLM research pipelines generate streams of evidence—retrieved literature, computational results, agent opinions—that frequently conflict. Existing aggregation strategies (majority vote, Elo tournaments, single-judge selection) either ignore conflicts entirely or resolve them by rhetorical weight, and both are vulnerable to two under-addressed failure modes: false convictions of well-formed claims built on low-quality evidence, and infrastructure failures (citation-verification network outages) silently polluting scientific verdicts. We present ConsensusCourt, an arbitration framework that borrows its procedural guarantees from the courtroom rather than from voting theory: every claim is presumed innocent until a detected *contradiction pair* convicts it (an asymmetric burden of proof, ABP); evidence is weighted by an explicit source-pedigree, recency, and relevance scorer whose confidence channel is structurally decoupled from conviction; and an escalation layer routes uncertain or unadjudicable cases to human review instead of guessing. Three results carry the paper's evidential weight and do not depend on benchmark authorship: (i) a counting law that calibrates the court's advantage over guilty-presumption screening—the log-log slope of false pairs against pairwise opportunities is 0.942 (expected 1.0) and the advantage-ratio slope is -0.857 (expected -1.0); (ii) a controlled network-failure experiment in which a three-state citation-verification policy holds the clean-claim false-positive rate at 0 under failure rates up to 100%, where the common two-state policy convicts up to 100% of innocent claims—the only variable being the failure-mapping policy over an otherwise identical verification stack; and (iii) a conviction–confidence decoupling invariant, preserved exactly under every scorer ablation. On the author-built synthetic layer of our four-layer 954-case benchmark (Ddb; evaluation at regression scale, §8), the full court reaches recall = 1.0 with FPR = 0.0 while majority vote, Elo calibration, and single-judge baselines fail on at least one side of the Pareto frontier at every threshold setting we scan. A per-case advantage decomposition identity holds exactly by construction, its empirical content carried by a pure layer attribution (residual = 0 on 148 negative-basis cases), and verdicts are order-invariant over 1,000 evidence permutations. The main benchmark and every red-line experiment are deterministic and zero-LLM; two registered LLM-side ablations (an LLM arbiter replacing the rule-based PI arbiter, and a PROClaim/DebateCV-style prosecution–skeptic–judge LLM debate protocol) are executed against a hosted LLM and reported in §6.12. All deterministic pipelines are guarded by pre-registered red-line assertions wired into a regression suite.

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1 Introduction

LLM-driven research agents now propose, execute, and report on entire scientific workflows. In drug-discovery pipelines, a single design decision may draw on dozens of evidence items produced by heterogeneous agents: literature retrievals (with citations that must resolve), structure-prediction outputs, simulation summaries, and free-text agent opinions. When two such items disagree—a paper reports that an allosteric inhibitor is effective against a resistance mutation, another that monotherapy fails in vivo—the pipeline must decide *which claim survives*, and it must do so procedurally: retro-justified aggregation ("the loudest evidence wins") is precisely how plausible-but-wrong conclusions propagate.

Two failure modes dominate. First, **false conviction**: systems that pre-screen evidence with a guilty-presumption rule ("any malformed evidence rejects the case") reject half of all well-formed claims in our benchmark (FPR 0.5, Section 6.4), because the mere presence of a defect somewhere in the evidence pool is treated as an indictment of the claim. Second, **infrastructure pollution**: citation verifiers call network APIs, and when those APIs fail, a naive two-state policy (valid vs. anything-else) classifies "could not verify" as "fabricated," convicting clean claims at rates that reach 1.0 under full network outage (Section 6.8). Neither failure mode is addressed in the evidence-aggregation channel by the multi-agent debate literature, which studies argumentation quality but not procedural burden of proof—even the recent courtroom- and debate-style verifiers [1, 2] resolve conflicts through LLM rhetoric rather than an explicit, auditable conviction rule; nor by peer-review conflict analysis [3, 4], which adjudicates conflicts among *human* reviewers rather than among machine-generated evidence items inside a running pipeline.

We contribute ConsensusCourt, an arbitration framework with four procedurally grounded components:

1. **Evidence scoring with explicit pedigree** (§3.2): every evidence item receives a source-pedigree, recency, and relevance score, combined as overall $= 0.4\,\text{source} + 0.2\,\text{recency} + 0.4\,\text{relevance}$, with a registered weight table for 13 agent roles.
2. **Asymmetric burden of proof (ABP)** (§3.3): conviction requires a detected *contradiction pair*—two evidence items taking opposing stances on the same aspect—conditioned by aspect matching, an object-validity gate, and graded intensity; evidence is otherwise presumed innocent, and scoring is structurally decoupled from conviction.
3. **A seven-step court procedure** (§3.4): symmetric pre-screening (hard constraints only), scoring, ABP adjudication, PI arbitration, independent review, human-BOSS override, and escalation, with six explicit escalation clauses.
4. **Three-state citation verification** (§6.8): `valid` / `invalid` / `network_failed`, in which infrastructure failures carry evidentiary *warning* status and can never convict.

Our evaluation rests on Ddb, a four-layer Dispute Benchmark (§5) spanning 954 cases: 600 synthetic L1 cases, 88 curated pseudo-conflict negatives (L1.5), 10 real literature disputes with verified PubMed identifiers (L2), and 256 intensity-graded cases (L4). The main benchmark and every red-line experiment are deterministic, zero-LLM pipelines with pre-registered red-line assertions executed by a CI-wired regression suite; two registered LLM-side ablations additionally invoke a hosted LLM (§6.12). Every number in this paper is traceable to a machine-readable report.

Contributions. (i) A procedural arbitration framework whose conviction channel is structurally decoupled from evidence scoring, with a theory-grounded advantage identity (§4); (ii) a

four-layer benchmark with keyword-channel locking so that synthetic and real-dispute layers measure the same detection channel (§5); (iii) a controlled experiment on network-failure discrimination in citation verification—to our knowledge no prior work *quantifies* how verification outages propagate into downstream verdicts: industrial triage systems already deploy unexamined variants of the three-state policy [5], and our contribution is the controlled failure-rate/FPR curve that measures the pollution cost of the ubiquitous two-state policy (§6.8); (iv) deterministic, red-line-guarded, zero-LLM evaluation infrastructure in which component ablations, scorer ablations, order-permutation robustness, source-label noise, and role-weight calibration are all reproducible from a single regression script (§6); and (v) two executed LLM-side ablations—an LLM arbiter replacing the rule core's PI arbiter and a protocol-level LLM-debate verifier on identical evidence pools—measured against the frozen court outputs under the same operating points (§6.12).

2 Related Work

Hallucination detection and self-verification. Surveys [6] catalogue hallucination modes in generation; SelfCheckGPT [7] samples self-consistency signals to detect unsupported content, and Reflexion [8] and deliberation methods [9] improve factuality by iterative verbal feedback. These methods judge *single* outputs against internal consistency. ConsensusCourt addresses a complementary problem: when a pipeline's evidence *pool* contains genuine, source-grounded disagreement, internal consistency signals cannot resolve it; an explicit adversarial procedure must.

Multi-agent LLM systems. Multi-agent debate improves reasoning by having models critique each other [10]; frameworks such as AutoGen [11] and MetaGPT [12] orchestrate role-specialised agents; the AI co-scientist [13] coordinates hypothesis generation and review through Elo tournaments. Critically, these systems' final aggregation is either voting, tournament ranking, or a single judge. Section 6.2 shows all three aggregation families fail on at least one side of the recall/FPR frontier on Ddb, and Kapoor et al. [14] independently argue that agent evaluations are dominated by cost- and rigor-insensitive methodology—our red-line-guarded deterministic harness is a direct response.

Machine-side courtroom and debate-based claim verification. Closest in spirit to ConsensusCourt are three recent systems that stage claim verification as an explicit adversarial proceeding. PROClaim [1] casts verification as a courtroom debate between Plaintiff and Defense agents before a Judge, with Progressive RAG expanding the evidence pool mid-debate and heterogeneous multi-judge aggregation enforcing calibration; DebateCV [2] (WWW 2026) pairs two debater agents with a moderator adjudicator trained by Debate-SFT post-training; TRUST Agents [15] combines a claim decomposer, a Delphi-style multi-agent jury, and a logic-aware aggregator for fake-news verification. All three are LLM-driven, operate offline on standalone controversial claims, and resolve conflicts by generated argumentation; ConsensusCourt is rule-driven, deterministic, and zero-LLM, adjudicating conflicts among machine-generated evidence items *at pipeline runtime* under an explicit burden of proof. The two approaches are complementary rather than competing: a procedural court of the kind built here can wrap the outputs of an LLM debate as its evidence pool, supplying the auditable conviction channel that stochastic argumentation lacks. We report a protocol-level version of this comparison—an LLM debate verifier with the PROClaim/DebateCV prosecution–skeptic–judge structure consuming identical evidence pools (§6.12); head-to-head evaluation against the third-party systems themselves remains future work (§8).

Adjudication of review conflicts. Closest to us, Cecil and Freedman [4] (MDHJudgments) study medical hallucination detection where *annotator* judgments conflict, and introduce dual-track adjudication protocols for label curation; Kumar et al. [3] (IMPACT/RevCI) analyse conflicts

among *human* peer reviews, proposing aspect-conditioned contradiction extraction and graded conflict intensity. Both adjudicate conflicts among human or labelling agents, offline, to curate datasets or triage AC decisions. ConsensusCourt adjudicates conflicts among machine-generated evidence items *at pipeline runtime*, where the decision is whether a research claim continues to execute. We adopt their aspect-conditioning and graded-intensity insights as design components (§3.3), and our dual-track escalation analysis (§3.4) mirrors their finding that adjudicators themselves diverge.

Citation verification and infrastructure failures. Citation-verification services are increasingly used as hallucination firewalls, and industrial systems have begun to implement three-way triage—*e.g.*, GPTZero's hallucination check routes unverifiable citations to an explicit "unknown" outcome rather than a fabrication verdict [5]—while abstention-aware evaluation is gaining ground on the research side of scientific citation generation [16]. What no prior work we are aware of *quantifies*, however, is how verification outages propagate into downstream verdicts: the industrial triage practices ship without a failure-rate/FPR curve, and citation-generation abstention metrics measure generation-side behaviour rather than downstream adjudication. Section 6.8 provides, to our knowledge, the first controlled false-positive curve comparing failure-handling policies under matched failure injection, showing the distinction is not implementation detail but a first-order scientific-validity property.

3 The ConsensusCourt Framework

3.1 Design principles

Three principles, each falsifiable in our experiments:

P1 (Presumption of innocence). A claim is rejected only when a detected contradiction pair convicts it. Absence of contradictions acquits, regardless of evidence quality. This inverts the guilty-presumption pre-screening common in retrieval firewalls, which we show costs 0.5 FPR on clean claims (§6.4).

P2 (Conviction–confidence decoupling). The scoring channel computes a *confidence* for acquittal semantics; the conviction channel is driven solely by detected contradiction structure. Any ablation of the scorer therefore cannot change recall/FPR—only calibrated confidence—an invariant we assert as a red line (§6.6).

P3 (Escalation over error). When the court cannot adjudicate— because evidence was filtered, confidence is low, or tracks diverge—it abstains and escalates to humans. Abstention is a correct answer, not a failure mode (§6.11, escalation-type case).

3.2 Evidence scoring with source pedigree

Each evidence item e carries a source key $s(e) \in \{$`pubmed`, `chembl`, `uniprot`, `experiment`, `agent_opinion`, …$\}$, an agent role $r(e)$ from the 13-role registry (Appendix A), a publication timestamp, and free-text content. The scorer produces

$$\text{overall}(e) \;=\; 0.4 \cdot \text{source}(e) \;+\; 0.2 \cdot \text{recency}(e) \;+\; 0.4 \cdot \text{relevance}(e), \tag{1}$$

where source combines the source key's pedigree weight with the agent role's registered weight, recency decays with evidence age, and relevance scores topical fit to the claim under review. The uniform-replacement value used in the source-pedigree ablation is the median of the eight registered source weights, 0.775.

3.3 Asymmetric burden of proof (ABP)

The rubric engine evaluates the case against a registry of contradiction keyword pairs (*e.g.*, "effective/ineffective," "promote/inhibit"), scoped by three mechanisms:

- **Aspect conditioning**: pairs only match within a shared biomedical aspect (safety, efficacy, mechanism, …), preventing cross-aspect pseudo-conflicts—the dominant failure mode predicted by Kumar et al. [3].
- **Object-validity gate**: ambiguous or non-co-referent object mentions block pair formation (*e.g.*, "drug X is toxic" vs. "drug Y is toxic" is not a contradiction about X).
- **Graded intensity**: each detected pair receives an intensity grade 1–3 determined by polarity, source-pedigree differential, and same-object corroboration; intensity 3 (adversarial-grade) triggers mandatory escalation (clause 5, Appendix B).

Same-source evidence pairs are skipped (a source cannot contradict itself), and a hard-constraint channel detects regulatory tool bans as single-evidence critical findings—*e.g.*, references to AlphaFold2/3 structure prediction, which the deployment platform prohibits system-wide (both AF2 and AF3 are banned; compliant chains such as OpenDDE/Chai1/Boltz are whitelisted). This channel operates independently of keyword pairs and is the only conviction path that survives ABP removal on synthetic data (§6.5). The ban covers AlphaFold2 [17] and AlphaFold3 [18].

3.4 The court procedure

`hold_court` executes seven steps: (1) symmetric pre-screening (hard constraints; *e.g.*, tool bans); (2) evidence scoring (Eq. 1); (3) ABP adjudication (contradiction pairs + intensity); (4) PI-agent arbitration over the surviving evidence, with a decision of *support*/*conditional support*/*reject* by weighted-average thresholds (≥ 0.7 / ≥ 0.4 / < 0.4); (5) independent review producing objections; (6) human-BOSS override item generation; (7) escalation determination by six clauses (Appendix B), including: any reviewer objection; any intensity-3 conviction; dual-track divergence (the rule track and the evidence-review track disagree in *direction*); and no-adjudicable- evidence abstention. Our headless experiment pipeline (`run_court_headless`) replicates this orchestration line by line and is locked to the production `hold_court` by a field-exact parity test on an in-memory database; any drift in production orchestration fails the suite.

4 Theoretical Guarantees

We state two structural propositions (set-algebra identities and implementation invariants—true by construction rather than empirically falsifiable) and one falsifiable theorem (the counting law), whose quantitative predictions drive the experimental design; full proofs and counting lemmas are in the project's outline document.

4.1 Setup

Let $C = \emptyset$ denote the negative basis (claims with no true contradiction). Define: D: the guilty-presumption pre-screen convicts (a defect exists in the pool); M: the ABP court convicts (a pseudo-contradiction pair fires); $S \subseteq \mathrm{D} \setminus \mathrm{M}$: defect-only cases (guilty screen fires, ABP does not); $M \backslash \mathrm{D}$: ABP-only false convictions. Under $C = \emptyset$ (no true contradiction), D and M events are by definition false positives of their respective mechanisms.

Table 1: DDB layers. "Channel" names the detection channel the layer is designed to measure. All layers are emitted by versioned generators and locked by consistency tests.

Layer	Size	Construction and target channel
L1 (synthetic disputes)	600	Balanced with/without-contradiction cases from controlled templates (300 defect-only, 150 clean-neutral within the negative half); measures ABP recall and FPR on synthetic ground truth.
L1.5 (pseudo-conflict negatives)	88	Three curated negative families: cross-aspect (40), ambiguous-reference (24), different-object-gate (24); measures gate purity (target FPR = 0).
L2 (real literature disputes)	10	First batch of hand-built disputes from real published literature; every evidence item carries verifiable PubMed identifiers, all verified against NCBI esummary (re-verified 2026-09-20, with a second-pass correction pass); gold contradiction pairs recorded; measures source-generalisation of ABP recall (H2b).
L4 (intensity-graded)	256	16 templates × 4 rounds × 2 Δ-buckets × 2 gate buckets with intensity gold labels from the S1–S5 decision table; measures grading fidelity.

Proposition 1 (Per-case advantage decomposition)**.** *On the negative basis $C = \emptyset$,*

$$\underbrace{\Pr(\mathrm{D}) - \Pr(\mathrm{M})}_{\text{advantage of ABP over guilty screen}} = \Pr(S) - \Pr(M \backslash \mathrm{D}). \tag{2}$$

The advantage equals defect-only cases correctly spared minus pseudo-conflict cases wrongly convicted. This is a set-algebra identity that holds by construction; its empirical content is the layer attribution measured in Section 6.9, not the closure itself.

Theorem 1 (Counting bound on the FPR ratio)**.** *Under a null case with m evidence items, single-claim defect rate p_0, per-pair pseudo-mismatch rate q_K, and pairwise dependency κ (Section 6.9 for the measured estimator), the expected FPRs satisfy*

$$\frac{\mathrm{FPR}_{guilty}}{\mathrm{FPR}_{ABP}} \approx \frac{1-(1-p_0)^m}{q_K\,\kappa\binom{m}{2}}, \qquad q_K \text{ through } K \sim \kappa\, q_K^2 \binom{m}{2} \text{ joint events,}$$

and the Bonferroni-consistent bound FPR *ratio* $\geq 1/(\kappa\binom{m}{2}\cdot\frac{q_K}{p_0})$ *tightens as m grows: the ABP advantage widens with case size, with crossover m^* where the counting-based budgets of the two mechanisms intersect.*

Proposition 2 (Order invariance)**.** *The canonical seven-step court produces a verdict invariant to the input evidence permutation: the verdict signature (passed, convicted-pair multiset, pending pairs) is identical for all orderings, because every aggregation is order-agnostic (global scoring, set-based pair detection). Streamed early-termination variants forfeit this guarantee (Section 6.7).*

5 The Dispute Benchmark (Ddb)

DDB has four layers, all generated or curated deterministically from a single seed (42):

H2b channel lock. Because L1 is generated by the same authors as the detector, generalisation is a standing threat. On the first 10 L2 cases, the ABP keyword channel (direct rubric evaluation)

recovers at least one gold contradiction pair in all 10 positives, and the L2 wording deliberately uses the registry's keyword pairs to express genuinely opposed stances—real positions that happen to hit the keyword channel, never fabricated sentences. L2 output is exported to JSON deterministically from the code single-source-of-truth with file↔code deep-equality locks. At $n = 10$ positives, recall 1.0 has a Wilson 95% interval of $[0.72, 1.00]$; the L2 result is therefore reported at hypothesis-screening resolution only, and confirmatory interpretation awaits the planned L2 expansion (§8).

Honesty boundary. L2 ground truth is expert-adjudication-pending; the gold pairs are construction-time annotations, and the dual-track expert annotation protocol (three domain experts plus an independent evidence-verification track, with Krippendorff's α reported even if low, following 4) is specified but not yet executed (§8).

6 Experiments

6.1 Evaluation discipline

All red-line experiments are **deterministic and zero-LLM**: evidence scoring, pair detection, arbitration, and escalation are pure code; no LLM is invoked in any of these pipelines (the production platform's LLM stack—vLLM-served Qwen [19]—is outside the scope of these experiments). Two registered ablations carve out of this discipline and do invoke a hosted LLM—the LLM-arbiter ablation and the protocol-level LLM debate of §6.12; they are excluded from the zero-LLM regression and ship with their own structural red lines. Each experiment asserts **pre-registered red lines** (*e.g.*, "ablation FPR must not exceed full-court FPR"), violations abort with non-zero exit codes wired into a regression suite; each check has a paired anti-test that verifies the checker itself catches fabricated violations. Every report is machine-readable JSON; Appendix D lists commands, and DATA_PROVENANCE.md maps every table in this paper to its report file.

Why zero-LLM for the core experiments. The deterministic, zero-LLM discipline is a methodological choice rather than a rejection of LLMs, for three reasons. (1) Variable isolation: the paper's causal claim is that the *ABP architecture* controls false convictions, not that a particular model is capable. Running the core experiments without an LLM purifies the architectural effect—any LLM component would confound attribution between "ABP works" and "the LLM is good." (2) Red-line verifiability: pre-registered assertions such as "ablation FPR must not exceed full-court FPR" require deterministic outputs; stochastic LLM sampling would make red-line violations undecidable. (3) Reproducibility: every number in Tables 2–5 is reproducible from a single regression script with no network or API key. The two LLM ablations of §6.12 deliberately break this discipline to test the complementary question—what happens when an LLM arbitrates *without* the ABP conviction rule—and the answer (FPR 0.500 for the single-LLM arbiter) confirms that the architectural invariant, not model competence, is what bounds false convictions.

Positioning within the agent-evaluation debate. This harness is a deliberate answer to the agent-evaluation critique: Kapoor et al. [14] show that agent benchmarks are dominated by cost- and rigour-insensitive methodology; REDAgentBench [20] demonstrates that evaluation verdicts can conflate actual violations with mere evidence visibility—a confusion isomorphic to the infrastructure-failure pollution studied in §6.8; and AtomCite [21] shows that noise in automatic (LLM-judge) annotation labels can bias measured verifier accuracy and reverse system rankings. These findings motivate DDB's deterministic, versioned, code-locked construction: primary labels are emitted by generators and locked by consistency tests, with expert adjudication planned as a separate validation layer rather than the primary label source (§5, §8).

Table 2: Main comparison on DDB L1+L1.5 ($n = 208$, seed 42). "Reject" is the primary operating point; "non-support" (reject or conditional) is a sensitivity operating point. Rejection-point Recall and FPR carry Wilson score 95% confidence intervals. Best per column in bold.

	Reject operating point						Non-support sensitivity		
System	Recall	95% CI	FPR	95% CI	Acc	Prec	Recall	FPR	Acc
CONSENSUSCOURT (full)	**1.000**	[0.940, 1.000]	**0.000**	[0.000, 0.025]	**1.000**	**1.000**	1.000	**0.000**	**1.000**
MajorityVote	0.550	[0.425, 0.669]	0.878	[0.816, 0.922]	0.245	0.203	0.883	0.993	0.260
EloArbiter	0.050	[0.017, 0.137]	0.628	[0.548, 0.702]	0.279	0.031	1.000	1.000	0.289
SingleJudge	0.000	[0.000, 0.060]	0.074	[0.042, 0.128]	0.659	0.000	0.833	0.939	0.284

6.2 Main comparison against aggregation baselines

Four systems consume the *same* EvidenceScorer scores on the same 208 cases (60 L1 positives + 148 negatives: 60 L1 + 88 L1.5); the only variable is the arbitration logic. Baselines: **MajorityVote** (equal-weight agent votes, thresholds 0.7/0.4), **EloArbiter** (role-weighted average calibrated through a single-round Elo transform, approximating tournament review 13), **SingleJudge** (max-confidence evidence decides alone).

Three findings. (1) **Constructive failure of vote-style aggregation**: majority vote rejects 87.8% of negatives—because the negative half deliberately contains defect-bearing cases, and low evidence quality drives rejection votes. (2) **Single-judge zero recall**: the highest-scoring evidence on genuine disputes is frequently on the wrong side; the judge never sees the opposing item. (3) **Sensitivity collapse**: under the "non-support" operating point, all three baselines push FPR to 0.94–1.0: without a contradiction channel there is no principled boundary between "conditional support" and "reject." The full court is the only system occupying recall = 1.0 / FPR = 0.0 on both operating points. The Wilson intervals make the separation unambiguous rather than point-estimate-lucky: the court's recall interval $[0.940, 1.000]$ and FPR interval $[0.000, 0.025]$ overlap with no baseline's interval in either column.

6.3 Threshold-sensitivity scan

A fair objection to Table 2 is that baselines are evaluated at a single, possibly unfavourable, operating point (support threshold 0.7, reject threshold 0.4). We therefore scan the full threshold grid $(t_{\text{sup}}, t_{\text{rej}}) \in \{0.3, \ldots, 0.9\}^2$—49 points—for each baseline, by *decision-primitive replay*: each case's scores and each baseline's decision primitive (vote counts, Elo role-weighted average, single best score) are computed once, and every grid point only remaps the primitive to a verdict. Production code is untouched; at the default thresholds the replayed metrics are red-line-locked to Table 2 field-for-field. Figure 1 plots the resulting (FPR, recall) Pareto frontier of each baseline against the full court. The empirical answer is unambiguous: **no baseline threshold point reaches the court profile** (recall = 1.0 at FPR = 0.0). At FPR ≤ 0.034 no baseline attains positive recall at all; the first positive-recall frontier point already costs FPR ≥ 0.43 (majority 0.432 at recall 0.033), and recall at or above 0.467 (elo) is only reachable at FPR ≥ 0.946. The court's advantage is thus a property of its contradiction-channel architecture, not of a threshold choice made on the baselines' behalf.

6.4 Pilot: ABP vs. guilty-presumption pre-screening

On the full 600-case L1 layer, the guilty-presumption pre-screen (any formal defect ⇒ case rejection, the standard retrieval-firewall policy) achieves recall 0.6 at FPR 0.5 overall—convicting 100% of

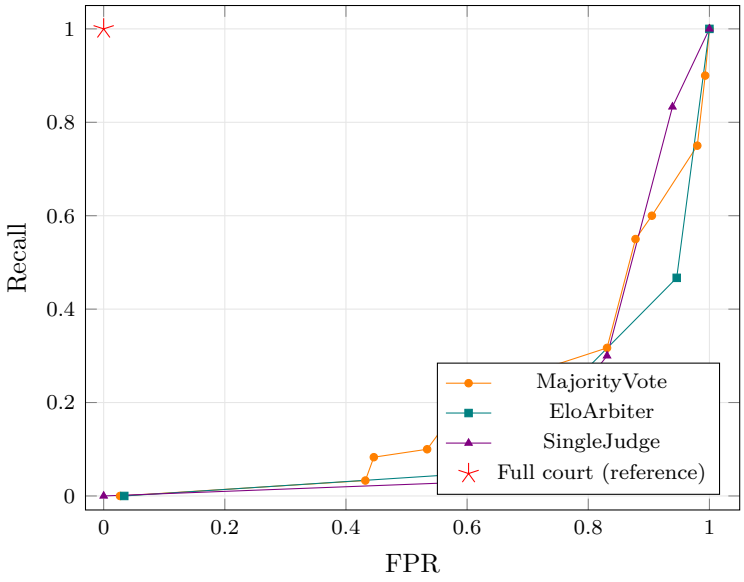


Figure 1: Pareto frontiers of the three aggregation baselines over the 7×7 threshold grid ($t_{\text{sup}}, t_{\text{rej}} \in \{0.3, \ldots, 0.9\}$, reject operating point), from decision-primitive replay with zero production-code changes. The star marks the full court, which is threshold-free by construction; no baseline grid point attains recall = 1.0 at FPR = 0.0. Data: `baseline_sensitivity_report.json`.

Table 3: Court component ablation on 60×2 L1 (seed 42). Conviction layer vs. escalation layer. no_abp removes the pairwise rubric (hard constraints remain); no_reviewer / no_boss remove the downstream review/override components.

	Conviction		Escalation	
Mode	Recall	FPR	Recall	FPR
court_full	1.000	0.000	1.000	1.000
no_abp	0.000	0.000	1.000	1.000
no_reviewer	1.000	0.000	1.000	1.000
no_boss	1.000	0.000	0.733	0.800

defect-only negatives (by design) but also 50% of clean-neutral negatives via collateral pollution. ABP holds recall 1.0 / FPR 0.0 on the same cases. This is the empirical form of Proposition 1: the guilty screen's $P(\text{D}) = 0.203$ on the 148-case negative basis is entirely defect-only (S), all of which ABP spares ($P(\text{M}) = 0$).

6.5 Component ablations

ABP is load-bearing: without the pairwise rubric, conviction recall collapses $1.0 \to 0.0$ (the synthetic layer contains no tool-ban content, so the hard-constraint channel—the only remaining conviction path—never fires). **Reviewer and BOSS are downstream of conviction** by design: removing them leaves conviction identical (a structural invariant asserted by red line). **BOSS is the primary escalation signal source**: removal drops escalation recall to 0.733 while conviction is untouched.

Escalation is not conviction. On the full court, escalation FPR is 1.0 with precision 0.5 (and 0.0 on the L1.5 basis): every defect-bearing-but-innocent case (the S set of Proposition 1) is

Table 4: Scorer-level ablation (60×2 L1, seed 42). avg_pass-conf = mean overall score of passed cases. Recall/FPR identical across rows (decoupling invariant); the observable effect is confidence inflation.

Configuration	avg. pass conf.	Interpretation
full_v03 (reference)	0.4994	calibrated baseline
no_source_pedigree ($s \to 0.775$)	0.5248	low-pedigree sources lifted
no_recency ($\to 1.0$)	0.6250	stale evidence conflated with fresh

routed to human review rather than auto-rejected. This is the intended semantics—escalation is the court's abstention channel, and measuring precision@escalation is how the routing budget is tuned (§6.11 shows the complementary escalation- type case where abstention is provably correct).

6.6 Scorer-level ablations

The conviction–confidence decoupling (P2) makes a scorer ablation's effects observable only through the confidence channel. We therefore run two scorer ablations with the full court's switches frozen, post-processing scores experiment-side (production code untouched), with two red-line layers: R1 (monotonicity: ablated FPR must not exceed and recall must not fall below full court) and R2 (effect direction: `no_recency` confidence must not decrease; `no_source_pedigree` confidence must differ from the original—proving the ablation is live).

Both red lines hold: recall/FPR are exactly 1.0/0.0 in all rows, and the confidence ordering quantifies each mechanism's contribution to calibration (+0.025 from source pedigree, +0.126 from recency—the largest single effect, showing recency decay is doing most of the work in discounting stale literature).

6.7 Robustness: order, size, streaming, and label noise

Order invariance (Proposition 2). 1,000 random permutations across 60 L1 positives, both full-court and baseline configurations: **0** divergent verdict signatures (invariant rate 1.0). The court's verdicts are a function of the evidence *set*, not its order.

Streaming ablation. A rule-based streamed variant that terminates on first contradiction (k=1) or first k-strong-streak (k=2) diverges from the canonical court in 70/1000 = 0.070 permutations (k=1; 10/60 cases affected; permuted recall 0.924 vs. canonical 0.95, court canonical 1.0) and 2/1000 = 0.002 (k=2, recall 0.998). Order sensitivity is real for early-termination designs and decays as the termination threshold tightens—supporting the design choice of full deliberation before disposition.

Counting-law calibration (Theorem 1). Under controlled null cases ($p_0 = 0.30$ designed defect injection, unique-source construction so that all $\binom{m}{2}$ pairs are live, 300 trials per $m \in \{3, 5, 10, 20\}$): the log-log slope of false pairs against $\binom{m}{2}$ is 0.942 (expected 1.0), the advantage-ratio slope against $(m-1)$ is -0.857 (expected -1.0), and the empirical ratio meets the Bonferroni bound pointwise at every grid point; the crossover is $m^* \approx 18.5$. The court's advantage widens with case size exactly as the counting law predicts.

Source-label noise (±20%). A corruption pass re-labels exactly $\max(1, \lfloor 0.2m \rceil)$ sources per case (50 replicates per case; only `source` fields change, with field-level oracle invariants asserted). On 7,400 negative replicates (L1 + L1.5), **zero** false convictions are induced—the only theoretical attack (un-ignoring a same-source skipped pair) has no fuel on these bases, and the red line treats this claim as falsifiable: any induced case would be written to an audit field with its pair endpoints

Table 5: Network-failure experiment (Fig. 2): FPR on clean claims / recall on invalid-reference claims, by policy and failure rate p.

	$p = 0$	$p = 0.3$	$p = 0.5$	$p = 1.0$
Three-state FPR (clean)	**0.000**	**0.000**	**0.000**	**0.000**
Two-state FPR (clean)	0.000	0.667	0.833	1.000
Three-state recall (invalid)	1.000	1.000	0.900	0.000[†]
Two-state recall (invalid)	1.000	1.000	1.000	1.000

[†]At full outage the three-state court abstains on all cases (refuses to adjudicate rather than guessing)—consistent with the presumption-of-innocence principle; the two-state policy's "perfect recall" is purchased entirely with clean-claim false positives.

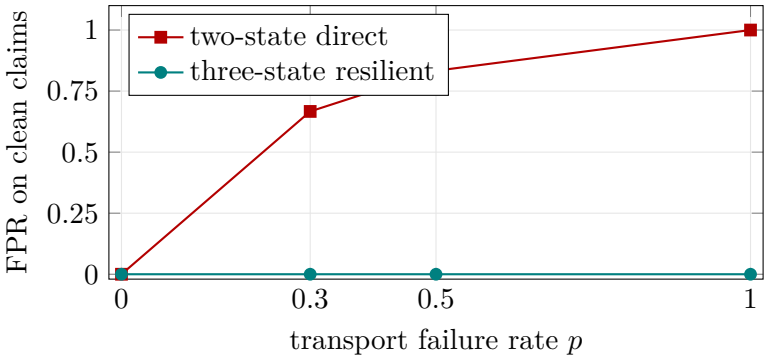


Figure 2: Clean-claim false-positive rate under injected citation-verification failures. The two-state policy's FPR is monotone in p and reaches 1.0 at full outage; the three-state policy is structurally flat at 0 (network failures cannot enter the conviction channel). Data: Table 5; source report `tests/ddb/network_failure_report.json`.

(observed: empty). On 3,000 positive replicates, 62 convictions are lost (2.07%; the sole channel is a core pair re-labelled same-source), leaving noisy recall 0.9793; pair-set flip rate 13.9%, intensity flip 14.1%, mean $|\Delta\text{conf}|$ 0.0076 (max 0.578, attained when all pairs are simultaneously suppressed). The positive-loss asymmetry is reported, not red-lined: it is a directional finding, not a defect.

6.8 Network-failure discrimination

Citation verifiers call network APIs; verification outages are routine. The community-standard **two-state** policy (`valid` vs. "anything else") conflates "could not verify" with "fabricated." Our production **three-state** policy maps `network_failed` to a warning that cannot convict, with 3× exponential-backoff retries and a cross-case TTL cache. We inject deterministic transport failures (`httpx.MockTransport`) at rates $p \in \{0, 0.3, 0.5, 1.0\}$ over 60 clean-claim cases (all references valid, shared 40-hot-PMID pool) and 20 invalid-reference cases (recall control). Both policies consume the identical production verification/retry/cache stack; the only variable is the failure-mapping policy.

Process metrics confirm the mechanisms: the three-state policy's mean attempts rise $1.00 \to 1.33 \to 1.75 \to 3.00$ with p (retry effect observable), and the cross-case cache absorbs 133 lookups (shared-pool reuse), cutting transport requests at $p = 0.3$ from 212 to 103. Six pre-registered red lines pass, including bit-exact determinism of a double-run at $p = 0.5$.

Relation to abstention-aware evaluation. The three-state policy instantiates the same design stance as abstention-aware evaluation in scientific citation generation, where hallucination-

Table 6: Proposition 1 calibration. Cell counts and closure of $\Pr(\mathrm{D}) - \Pr(\mathrm{M}) = \Pr(S) - \Pr(M \backslash \mathrm{D})$ on the negative basis.

Config	$P(\mathrm{D})$	$P(\mathrm{M})$	$\Pr(S)$	$\Pr(M \backslash \mathrm{D})$	residual
full_v03	0.2027	0.0000	0.2027	0.0000	**0**
baseline_v02	0.2027	0.5946	0.2027	0.5946	**0**

rate-centred metrics dominate [16]: uncertainty should yield an explicit non-committal outcome, not a conviction. The contribution here is orthogonal—citation-generation abstention metrics measure generation-side behaviour, whereas Table 5 quantifies the downstream adjudication cost of omitting the third state, as a function of the transport failure rate.

6.9 Advantage-decomposition calibration

On the 148-case negative basis ($C = \emptyset$; 60 L1 negatives of which 30 defect-only, plus 88 L1.5 pseudo-conflicts), with the real pre-screen as the D labeler and the real ABP rubric as the M labeler:

The identity holds *by construction* (residual 0, tolerance 10^{-12}, for both configurations—this is expected, since the identity is a set-algebra proposition, not an empirical finding); the empirical content is the pure layer attribution: all S cases (30/30) come from L1 defect-only; all $M \backslash \mathrm{D}$ (88/88 under baseline switches) come from L1.5 pseudo-conflicts—each benchmark family provably measures exactly the channel it was built for.

Dependency calibration. On controlled null cases ($m = 20$, 300 trials, 57,000 pair opportunities per configuration): baseline $q_K = 0.0355$ vs. full $q_K = 0.0058$ (the gates cut per-pair mismatch rates 6–13×); adjacent-pair joint events yield $\kappa = 1.42$ (baseline) and 2.47 (full), both above the $\kappa \geq 1$ independence line (stronger dependence in the full court is expected—gates induce correlated suppression), with joint-event counts (1,829 / 86) adequate for estimation.

6.10 Role-weight calibration

The 13 agent roles' evidence weights (Appendix A) were calibrated by single-role perturbation over a 208-case pool (L1 60×2 + L1.5), for both arbitration arbiters that consume role weights. Red lines: coverage ≥ 5 cases/role (achieved 27–55), positive control (×0.1 perturbation) must flip ≥ 1 decision (achieved 7/6/6 flips on pi-agent/drug-researcher/report-expert—proving the pipeline is live), and decision flip rate ≤ 0.05 at ±20%.

Observed maximum flip rate: $4/208 = 0.0192$ (data-specialist at ×0.8, identical for both arbiters), with flips occurring only in the down-weight direction and far below the 0.05 line. Verdicts are driven by contradiction structure, not by weight tuning: the role table is a calibration surface, not a decision surface.

6.11 End-to-end case studies on real literature disputes

Two L2 disputes, run through the full seven-step court with full intermediate-state logging (timeline, ABP pair ledger with intensities, per-evidence scores, filtered-evidence audit, verdict chain):

Deadlock type (DDB-L2-0001): EGFR triple-mutation inhibition. Both sides carry top-venue evidence (allosteric route: Nature 2016 [22] / Cancer Lett 2017; ATP-covalent route: NEJM FLAURA/OS [23]). Zero evidence is filtered; ABP detects the gold contradiction pair (L2-01-EV-02 × L2-01-EV-03, intensity 3), the case is convicted, and the PI arbitrates *conditional*

System	Recall ↑	95% CI	FPR ↓	95% CI
Court (frozen reference, rule-based PI arbiter)				
ConsensusCourt	1.000	[0.940, 1.000]	0.000	[0.000, 0.025]
LLM arbiter (208 cases, 0 unresolved failures)				
LLM arbiter	0.983	[0.911, 0.997]	0.500	[0.421, 0.580]
LLM debate (208 cases × 3 roles, 0 unresolved failures)				
LLM debate	0.700	[0.575, 0.801]	0.372	[0.298, 0.452]

Table 7: Executed LLM-side ablations on the 208-case pool (reject operating point; positive class = case contains a true contradiction). The court row is frozen from the pre-registered baseline comparison; the two LLM rows are executed results with resume-for-missing-calls (each case retains its first successful response; single model, one protocol round per case). Sensitivity (non-support) operating points, per-verdict distributions, and call-failure accounting are in the accompanying JSON reports.

support at confidence 0.528; four escalation clauses fire (low confidence, conditional-support policy, reviewer objection, intensity-3 mandatory review). This is the "both sides are credible" regime in which vote-style aggregation degenerates: the court produces a traceable chain from detected pair to human-review routing.

Escalation type (DDB-L2-0005): AF3 complex-prediction dispute. Every evidence item cites AlphaFold2/3, hitting the platform's structure-prediction tool ban; the symmetric pre-screen filters all evidence; the court finds itself with no adjudicable evidence, abstains (decision *unable to adjudicate*, confidence 0.0), and escalates the entire case to human review. Gold-pair detection is not applicable in this regime—the gap between the two cases' regimes is itself a finding: the court distinguishes "conflict to resolve" from "nothing to judge," and refuses to fabricate a verdict in the latter.

Honesty note. These case studies are grounded in verified real literature but are *not* production-platform runs; the outline's envisioned "5-round group-chat deadlock" narrative segment is replaced, faithfully, by the seven-step court timeline, and production-run case studies (ASFV/canine-vaccine pipelines) await the L3 layer (§8).

6.12 Executed LLM-side ablations

The two ablations registered as pending in the research plan—swapping the rule-based PI arbiter for a hosted LLM, and running a PROClaim/DebateCV-style prosecution–skeptic–judge debate verifier on identical evidence pools—have now been executed against a hosted LLM (`qwen-flash`, temperature 0; configuration and commands in Appendix D). Both are evaluated on the same 208-case pool as Table 2 (60 positives, 148 negatives), with the court's outputs *frozen in advance* from the pre-registered baseline report, under the identical reject and non-support operating points. Evidence is presented blind: `evidence_id`, `source`, `role`, `content` only—no scores, no gold labels, no court verdicts. Table 7 reports the outcome, taken verbatim from `llm_arbiter_report.json` and `llm_debate_report.json`.

Three observations, reported regardless of direction. **(1) Recall is not what separates the rule core from LLM arbitration.** The single-LLM arbiter reaches 0.983 recall—within one positive case of the court's 1.0—because on this synthetic benchmark outright contradictions are linguistically conspicuous. **(2) The separation is entirely in the false-conviction dimension.** The LLM arbiter convicts FPR = 0.500 [0.421, 0.580] of the 148 well-formed negatives while the

court convicts none (FPR 0.000, CI upper bound 0.025); on the non-support operating point the arbiter convicts 0.980 of negatives (accuracy 0.303). The asymmetric burden of proof—not rhetorical competence—is what keeps benign evidence out of the conviction column, which is precisely the paper's central claim measured on the LLM's home turf. **(3) Adding adversarial structure without a conviction rule does not help.** The 3-call debate protocol *degrades* relative to single-shot arbitration (recall 0.700 vs. 0.983; Δrecall $= -0.300$ vs. the frozen court) and never supports a well-formed claim (0 support verdicts out of 208—consistent with the prosecution's framing prevailing over the skeptic's across both polarities, as judged by the final verdict distribution), while still convicting 0.372 of negatives. Structured LLM argumentation inherits the same false-conviction failure mode that motivates the court's presumption of innocence, and the debate framing adds no mechanism to bound it.

Honesty note. These runs are exploratory rather than tuned: one hosted model at temperature 0, one round each, no prompt ensembling. A pre-registered configuration deviation is declared: the planned `max_tokens`=300 was raised to 2048 because the hosted `qwen-flash` gateway enables chain-of-thought by default and the 300-token budget was exhausted by reasoning tokens, leaving the verdict payload empty; the deviation and its reason are recorded in the reports' `max_tokens_deviation` fields. Structural red lines all hold (complete response coverage, verdict-enum integrity with raw-response re-parsing, frozen court reference match); conservative handling of unresolved cases (not-caught under reject, caught under non-support) is declared in the reports' `prediction_mapping` fields. Results follow the same pre-registration discipline as the deterministic experiments and are excluded from the zero-LLM regression.

7 Discussion

Why courtroom procedure, not voting. The baseline failures in Table 2 are not implementation artefacts but category errors: majority voting aggregates *preferences*, while scientific arbitration must aggregate *evidence under a burden of proof*. Once conviction requires a contradiction pair, the system's recall/FPR profile is governed by detection quality (gates, aspects, intensity) rather than by vote thresholds—and every detection component is individually ablatable (Tables 3–4) with red-line-guarded attribution.

Escalation as a first-class outcome. The escalation layer's apparently poor "precision" (0.5 on smoke sets, 0.0 on the negative basis) is the design working as intended: the S set of Proposition 1—defect-bearing, contradiction-free, therefore innocent—*must* be routed to humans rather than auto-rejected. The operational question is routing budget, not escalation FPR; the deadlock / escalation case-study pair (§6.11) illustrates both regimes.

An escalation cost model. The routing question can be made falsifiable. Let c be the cost of one human review, L the loss incurred by one wrongful automatic rejection of an innocent claim, N_{esc} the number of escalated cases, and $\hat{p}$ the precision@escalation. The escalate-over-error policy costs $C_{\text{esc}} = c\,N_{\text{esc}}$; an auto-reject policy that dismisses exactly the escalated set costs $C_{\text{auto}} = (1-\hat{p})\,L\,N_{\text{esc}}$. Escalation is the cheaper policy if and only if $c/L < 1-\hat{p}$. On the 148-case negative basis (§6.9), every escalated case is innocent ($\hat{p} = 0$), so escalation dominates for any $c < L$: one wrongful auto-rejection must cost more than one review. On the 60×2 L1 smoke set (§6.5), $\hat{p} = 0.5$, so escalation dominates iff $c < L/2$. The model turns the P3 principle from a design stance into a testable proposition: any deployment reporting precision@escalation and review cost can read off the cheaper channel, and the red-line-guarded precision measurements (§6.5, §6.9) supply the measured inputs.

Infrastructure failures are a scientific-validity concern. Table 5 shows that a standard

engineering default (fail closed) translates directly into convicting up to 100% of clean claims during outages. A three-state evidentiary policy costs nothing at $p = 0$, preserves most recall at moderate p through retries and caching, and abstains rather than fabricates at full outage. We suggest that any verifier-gated adjudication system report its failure-mode FPR curve alongside its clean-condition accuracy.

ABP and LLMs are complementary, not competing. A common misreading of the zero-LLM discipline is that ConsensusCourt rejects LLM integration. The opposite is true: ABP is a *procedural architecture* structurally decoupled from the arbitration substrate, and LLMs can occupy any of three slots inside it. (i) As the *scorer*: an LLM can replace the rule-based source-pedigree/relevance scorer (Eq. 1); by the P2 decoupling invariant this changes confidence calibration but cannot move recall or FPR. (ii) As the *contradiction detector*: an LLM can enrich the aspect-matching and intensity-grading that feed ABP conviction, potentially raising recall on semantically non-obvious disputes. (iii) As the *arbiter*: an LLM can replace the PI-agent in step 4 of §3.4; §6.12 tests this slot *without* ABP and finds FPR 0.500, which is precisely why the ABP conviction rule must be retained regardless of arbiter choice. The target production architecture is therefore *LLM components inside an ABP envelope*, not LLM-as-verifier without procedural guardrails; the zero-LLM core experiments isolate the envelope's contribution, and the LLM ablations quantify what happens when the envelope is removed.

8 Limitations

We state the boundaries of the present evidence explicitly.

1. **No user study yet.** The planned 10-participant blind A/B evaluation (SUS trust, NASA-TLX workload, open feedback vs. Elo and single-judge outputs) has not been run; all claims here concern system behaviour, not user experience.

2. **No production-run (L3) layer.** DDB's L3 layer—claims, evidence, and verdicts harvested from real platform runs, including the group-chat deadlock narratives and wet-lab action orders—does not exist yet. The case studies (§6.11) use verified real literature but are benchmark-grounded, not production logs.

3. **LLM-side ablations: partial.** Two of the registered LLM-side ablations have now been executed against a hosted LLM (qwen-flash, temperature 0): the LLM-arbiter ablation (swapping the rule-based PI arbiter for a direct LLM three-verdict arbiter) and a protocol-level LLM-debate verifier following the PROClaim [1] and DebateCV [2] prosecution–skeptic–judge structure, both consuming identical evidence pools with the court's outputs frozen in advance and evaluation under the same reject/non-support operating points of Table 2 (§6.12). The third registered ablation—replacing the position-weight proxy with LLM-based structural role annotation—remains pending, as does head-to-head evaluation against the third-party verifier systems themselves; the protocol-level comparison reported here uses our own debate prompts rather than the systems' trained pipelines.

4. **L2 scale and annotation status.** The L2 layer is 10 cases (target 45+), all positives (FPR-relevant neutral sets are a second-batch item), and expert adjudication is pending; gold pairs are construction-time annotations, and keyword-channel sensitivity means ABP's L2 conclusions must eventually be cross-validated with expert and LLM-judge channels. The expansion protocol is already designed: grow L2 to at least 45 cases including a neutral negative family (target FPR = 0 under the same gate-purity red line as L1.5), have every case annotated

by three domain experts under the dual-track protocol (three experts plus an independent evidence-verification track), and report Krippendorff's α even if low, following 4; the pre-registered acceptance red line is that ABP recall on expert-adjudicated L2 must stay within the Wilson interval reported in §5. Results below that bound would falsify the generalisation claim and will be reported as such.

5. **Third-party benchmark cross-validation.** All current benchmark layers are author-built. The next cycle adds a cross-validation arm on at least one independently constructed claim-verification benchmark, with the court and the strongest baseline evaluated under identical splits, prompts (where applicable), and operating points; divergence between author-built and third-party results will be analysed and reported regardless of direction. This arm is designed but not yet executed, and no third-party numbers appear in this paper.

6. **Smoke-scale statistics.** Reported experiments run at regression scale (60–300 cases per cell) for determinism and CI cost; the pre-registered main experiment (600×2 full-combinatorial ablations with significance tests) runs on the same pipelines at scale.

7. **Zero-LLM scope.** The main benchmark, all baselines, and all red-line experiments are deterministic rule systems; the two LLM-side ablations of §6.12 are the only pipelines that invoke an LLM, and the production platform's LLM components (evidence authoring, group-chat debate) remain upstream and unevaluated here.

9 Conclusion

ConsensusCourt reframes evidence aggregation in multi-agent research pipelines as procedural arbitration: presume innocence, convict only on detected contradiction structure, score for calibrated confidence, escalate what cannot be judged, and never let infrastructure failures impersonate scientific findings. Three results carry the evidential weight and do not depend on benchmark authorship: the counting law's calibrated advantage (slopes $0.942/-0.857$ against expected $1.0/-1.0$, with the empirical ratio meeting the Bonferroni bound at every grid point), the three-state verification policy's zero network-failure false positives under matched injection, and the conviction–confidence decoupling invariant preserved under every scorer ablation. On the author-built synthetic layer of the 954-case benchmark (evaluation at regression scale, §8), the full court is the only system that holds recall 1.0 / FPR 0.0 against all three standard aggregation baselines at every threshold setting scanned, and its verdicts are permutation-invariant. Every number from the deterministic pipelines is produced under pre-registered red lines wired into CI, and the two executed LLM-side ablations report their hosted-LLM results against the same frozen operating points (§6.12). The path to a complete preprint runs through the pending items above: expansion and expert adjudication of L2, third-party benchmark cross-validation and third-party head-to-head verifier evaluation, the L3 production layer, the user study, and the LLM-based structural role annotation ablation—each is a bounded, scripted increment on infrastructure that already exists.

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A Agent role evidence-weight registry

Evidence-credibility weights used by both arbiters (registered per role; unregistered roles fall back to 0.5; the platform's `data-analyst` implements the paper's `data-specialist` profile, with the platform-side fallback difference disclosed in the project's role-profile appendix):

Design principles: first-hand domain evidence > computational heuristics > organisational/expressive evidence; adjacent gap ≥ 0.05.

Role	Weight	Role	Weight
pi-agent	1.0	data-specialist	0.75
drug-researcher	0.9	qc-inspector	0.7
structural-biologist	0.9	knowledge-engineer	0.65
computational-chemist	0.85	reflexion-agent	0.6
synthetic-chemist	0.8	project-coordinator	0.6
antibody-engineer	0.8	report-expert	0.5
peptide-designer	0.75		

B Escalation clauses

The escalation decision is the disjunction of: (1) any reviewer objection; (2) PI confidence below the conditional-support threshold; (3) dual-track divergence between the rule-arbitration track and the evidence-review track; (4) decision in a policy-mandated review category; (5) any intensity-3 convicted contradiction pair (mandatory human review); (6) insufficient adjudicable evidence (abstention). Clauses 4 and 6 are what make the escalation-type case study (§6.11) behave as it does: a fully filtered case cannot be convicted *or* acquitted, so it must be escalated.

C Pre-registered red lines (summary)

Each experiment family ships with machine-checked red lines, each paired with an anti-test that fabricates a violation to prove the checker is live: baseline comparison (court strictly dominates all three baselines on both operating points); component ablation (no_abp conviction collapse on L1; reviewer/BOSS conviction invariance; BOSS escalation monotonicity; headless pipeline parity with production `hold_court`); scorer ablation (R1 monotonicity; R2 effect direction); robustness (permutation divergence $= 0$; counting-law slopes within bands; noise-induced negative convictions $= 0$ with audited empty set; noise recall ≥ 0.95; mean $|\Delta\text{conf}| \leq 0.05$); advantage decomposition (exact closure; zero critical evidence; $P(S) > P(M\backslash\text{D})$; q_K ordering; $\kappa \geq 1$); network failure (three-state FPR $\equiv 0$; recall profile; two-state monotone pollution curve; two-state recall invariance; bit-exact double-run; cache effect); role calibration (coverage, positive control, flip rate).

D Reproducibility

All experiments run from a single regression entry point, the bash script `backend/tests/scripts/run_v03_regression.sh`, with deterministic seed 42 and no network or LLM calls (the network-failure experiment's failures are injected at the transport layer). The two LLM-side ablations of §6.12 run outside this regression via a dedicated entry point and require network access to a hosted LLM endpoint. Per-experiment commands and report files (last two rows: network required):

The last two rows require network access to a hosted LLM (config via `LLM_EXPERIMENT_*` environment variables; model `qwen-flash`, temperature 0) and are excluded from the zero-LLM regression.

DDB layer files: `tests/ddb/{l1,l15,l2,l4}_cases.json`, each locked to its generator by deep-equality consistency tests.

Experiment	Script	Report
Main comparison	`run_baseline_comparison.py`	`baseline_comparison_report.json`
Threshold sensitivity	`run_baseline_sensitivity.py`	`baseline_sensitivity_report.json`
Component ablation	`run_ablation_v03.py --court`	`ablation_v03_report.json`
Scorer ablation	`run_ablation_v03.py --scorer-ablation`	`ablation_v03_report.json`
Robustness A–D	`run_robustness_experiments.py`	`robustness_report.json`
Decomposition	`run_advantage_decomposition.py`	`advantage_decomposition_report.json`
Role calibration	`run_role_weight_calibration.py`	`role_weight_calibration.json`
Network failure	`run_network_failure_experiment.py`	`network_failure_report.json`
Case studies	`run_case_study.py`	`case_study_report.json`
H2a pilot	`pilot scripts (tests/ddb)`	`pilot_h2a_report.json`
LLM arbiter ablation	`run_llm_arbiter_experiment.py`	`llm_arbiter_report.json`
LLM debate (protocol-level)	`run_llm_debate_experiment.py`	`llm_debate_report.json`